

AKKOL, Kazakhstan

WMO: 350850

Lat: 52.000N

Lon: 70.950E

Elev: 384

StdP: 96.80

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.8	-29.6	-35.8	0.1	-32.7	-32.6	0.2	-29.5	11.9	-11.1	10.3	-8.3	1.7	0	0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	31.1	17.5	29.2	17.2	27.4	16.6	19.4	27.1	18.6	26.1	17.9	24.9	3.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	12.6	21.8	16.1	12.0	21.1	15.2	11.3	20.7	57.1	27.0	54.5	26.2	52.0	25.0	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.9	8.4	7.4	-35.8	34.1	2.8	1.8	-37.8	35.4	-39.5	36.5	-41.0	37.5	-43.1	38.8		
			-35.9	21.0	2.8	1.4	-37.9	22.1	-39.6	22.9	-41.1	23.7	-43.2	24.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.5	-16.4	-14.9	-7.0	4.9	12.7	17.8	18.8	17.3	11.2	4.0	-6.0	-13.1
	DBStd	14.01	7.11	6.69	6.23	5.94	4.91	4.04	3.47	4.03	4.79	5.04	7.36	6.94
	HDD10.0	3682	819	696	527	171	29	2	0	2	41	196	482	717
	HDD18.3	5909	1077	929	785	403	183	57	37	67	217	446	732	976
	CDD10.0	954	0	0	0	18	112	236	272	229	78	8	0	0
	CDD18.3	140	0	0	0	0	8	41	51	36	4	0	0	0
	CDH23.3	1801	0	0	0	7	146	525	548	473	101	1	0	0
	CDH26.7	541	0	0	0	1	31	164	163	154	29	0	0	0
Wind	WSAvg	3.1	3.0	3.5	3.5	3.4	3.3	2.8	2.6	2.5	2.7	3.0	3.4	3.5
Precipitation	PrecAvg	372	21	17	22	26	34	38	68	38	23	31	27	27
	PrecMax	597	45	36	50	52	83	97	137	121	54	86	56	78
	PrecMin	219	5	5	5	8	3	5	12	12	2	4	14	6
	PrecStd	83	10	9	13	11	19	25	34	24	14	18	12	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.2	0.4	8.3	24.1	30.0	32.9	33.3	33.4	30.5	21.9	9.3	0.5
		MCWB	-0.4	-0.4	4.7	12.4	17.3	17.8	18.5	19.2	16.0	10.8	5.7	-0.5
	2%	DB	-1.8	-1.3	4.2	20.6	26.8	30.5	30.5	30.5	26.2	18.0	6.5	-0.6
		MCWB	-2.4	-2.2	2.2	11.4	14.5	16.9	18.1	17.6	14.0	9.4	4.4	-1.4
	5%	DB	-4.0	-2.9	2.5	18.0	24.7	28.5	28.3	28.2	22.7	15.2	4.2	-2.0
		MCWB	-4.6	-3.7	1.1	10.2	13.6	16.8	17.8	17.1	12.9	8.3	2.6	-2.7
	10%	DB	-5.9	-4.8	1.0	14.9	22.6	26.4	26.4	25.9	20.1	12.1	2.2	-3.7
		MCWB	-6.6	-5.6	-0.2	8.6	12.9	16.2	17.4	16.4	11.7	7.1	1.0	-4.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.4	-0.3	5.0	13.8	17.2	19.9	20.6	20.2	17.1	11.9	6.7	-0.2
		MCDB	0.1	0.2	8.4	21.6	25.8	28.8	28.0	27.9	27.8	18.5	7.8	0.2
	2%	WB	-2.3	-2.0	2.5	12.1	15.6	18.8	19.6	18.9	15.0	10.4	4.5	-1.4
		MCDB	-1.8	-1.3	4.0	19.5	24.1	27.1	26.9	27.0	23.1	16.1	6.3	-0.8
	5%	WB	-4.5	-3.7	1.2	10.5	14.5	17.9	18.9	18.0	13.6	8.8	2.7	-2.7
		MCDB	-4.0	-3.0	2.4	16.7	22.6	25.5	26.0	25.6	21.3	14.2	4.1	-2.1
	10%	WB	-6.5	-5.5	-0.1	9.0	13.5	17.0	18.2	17.2	12.4	7.4	1.1	-4.3
		MCDB	-5.9	-4.8	1.0	14.4	21.2	24.4	24.6	24.2	18.9	11.8	2.1	-3.7
Mean Daily Temperature Range	5% DB	MDBR	8.5	9.2	8.8	10.7	12.9	12.2	11.1	12.3	12.3	9.6	7.1	7.7
		MCDBR	7.8	7.9	7.4	16.0	16.5	15.9	15.1	16.2	17.3	15.7	8.3	8.0
		MCWBR	7.6	7.5	5.9	8.3	7.1	6.2	5.9	6.5	7.5	8.8	6.4	7.6
	5% WB	MCDBR	7.7	7.9	7.1	14.2	14.4	13.1	12.3	13.7	15.4	13.6	7.5	7.5
		MCWBR	7.6	7.7	6.0	8.1	7.1	6.1	5.6	6.5	7.2	8.2	6.2	7.4
Clear-Sky Solar Irradiance	taub	0.286	0.316	0.368	0.404	0.398	0.404	0.425	0.409	0.356	0.339	0.319	0.286	
	taud	2.066	2.026	1.971	2.140	2.235	2.288	2.224	2.250	2.352	2.302	2.144	2.050	
	Ebn at Noon	681	771	800	827	854	850	822	815	823	752	634	600	
	Edh at Noon	78	109	142	137	131	125	131	121	97	83	73	67	
All-Sky Solar Radiation	RadAvg	1.33	2.47	3.89	4.73	5.85	6.20	5.69	5.01	3.72	2.02	0.97	0.89	
	RadStd	0.10	0.09	0.38	0.43	0.52	0.49	0.42	0.35	0.26	0.26	0.12	0.12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page